INTERSECTIONS OF NORMAL SUBEXTENSIONS ARE NORMAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Gives the largest normal subextension inside any family intersection, used when constructing normal closures as intersections of splitting fields.

Editorial note. The label lemma-intersect-normal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter definition of normal: an algebraic extension E over F is normal when every irreducible polynomial over F with a root in E splits completely over E
- The intersection of a family of intermediate fields is an intermediate field

Lemma 0.1. *Let F be a field. Let M/F be an algebraic extension. Let $M/E_i/F$, $i \in I$ be subextensions with E_i/F normal. Then $\bigcap E_i$ is normal over F.*

Proof. Direct from the definitions. □

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